

Exp No:28

COUNTING SEMAPHORE IN SYSTEMC

```
anish@DESKTOP-BUFCRBD: ~  
GNU nano 8.7.1 /home/anish/systemc_test/counting_semaphore.cpp  
#include <systemc.h>  
  
SC_MODULE(counting_semaphore) {  
    int sem;  
  
    void task() {  
        while(true) {  
            wait(1, SC_SEC);  
  
            if(sem > 0) {  
                sem--;  
  
                cout << "Task Accessing Resource at "  
                    << sc_time_stamp()  
                    << " Remaining: " << sem << endl;  
  
                wait(2, SC_SEC);  
  
                sem++;  
            }  
        }  
    }  
  
    SC_CTOR(counting_semaphore) {  
        sem = 2;  
  
        SC_THREAD(task);  
        SC_THREAD(task);  
        SC_THREAD(task);  
    }  
};  
  
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^I Execute    ^C Location   ^U Undo       ^A Set Mark   ^] To Bracket  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line ^E Redo       ^- Copy      ^_ Where Was
```

```
anish@DESKTOP-BUFCRBD: ~  
GNU nano 8.7.1 /home/anish/systemc_test/CMakeLists.txt  
cmake_minimum_required(VERSION 3.10)  
  
project(CountingSemaphore)  
  
set(CMAKE_CXX_STANDARD 17)  
  
include_directories(/usr/include)  
  
add_executable(counting_semaphore counting_semaphore.cpp)  
  
target_link_libraries(counting_semaphore systemc)
```

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anish@DESKTOP-BUFCRBD: ~  
-- Check for working CXX compiler: /usr/bin/c++ - skipped  
-- Detecting CXX compile features  
-- Detecting CXX compile features - done  
-- Configuring done (3.1s)  
-- Generating done (0.0s)  
-- Build files have been written to: /home/anish/systemc_test/build  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ make  
[ 50%] Building CXX object CMakeFiles/counting_semaphore.dir/counting_semaphore.cpp.o  
[100%] Linking CXX executable counting_semaphore  
[100%] Built target counting_semaphore  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ ./counting_semaphore  
  
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46  
Copyright (c) 1996-2025 by all Contributors,  
ALL RIGHTS RESERVED  
  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0  
In file: kernel/sc_object_manager.cpp:153  
  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1  
In file: kernel/sc_object_manager.cpp:153  
Task Accessing Resource at 1 s Remaining: 1  
Task Accessing Resource at 1 s Remaining: 0  
Task Accessing Resource at 3 s Remaining: 0  
Task Accessing Resource at 4 s Remaining: 0  
Task Accessing Resource at 5 s Remaining: 0  
Task Accessing Resource at 6 s Remaining: 0  
Task Accessing Resource at 7 s Remaining: 0  
Task Accessing Resource at 8 s Remaining: 0  
Task Accessing Resource at 9 s Remaining: 0  
anish@DESKTOP-BUFCRBD:~/systemc_test/build$ |
```

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